

RECEIPTIQ PROJECT REPORT

Project Title:

ReceiptIQ – Privacy First Personal Finance Agent

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INTRODUCTION

ReceiptIQ is an AI-based personal finance assistant that extracts information from receipt images using OCR and categorizes expenses using machine learning.

The goal of this project is to automate expense tracking while maintaining user privacy by ensuring all processing is done locally without cloud services.

OBJECTIVE

- To extract text from receipt images using OCR
- To preprocess and structure extracted data
- To classify expenses into categories
- To provide analytics and insights

MY CONTRIBUTION

As a Data Engineer, my contributions include:

- Preparing and organizing receipt datasets
- Cleaning and preprocessing OCR-extracted text
- Handling noisy and unstructured data
- Designing data pipelines for smooth processing
- Ensuring compatibility between OCR output and ML input

METHODOLOGY

Step 1: Input receipt images

Step 2: OCR using EasyOCR

Step 3: Text cleaning and preprocessing

Step 4: Feature extraction using TF-IDF

Step 5: Classification using ML models

Step 6: Output analytics and visualization

TECHNOLOGIES USED

- Python
- EasyOCR

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- Scikit-learn
- Pandas & NumPy
- OpenCV
- Gradio

CHALLENGES FACED

- Handling low-quality receipt images
- Extracting structured data from noisy text
- Managing inconsistent formats of receipts
- Avoiding duplicate entries

LEARNINGS

- Practical use of OCR in real-world applications
- Importance of data preprocessing
- Working with unstructured datasets
- Building end-to-end ML pipelines

CONCLUSION

ReceiptIQ is a practical AI solution for personal finance tracking. It demonstrates the integration of OCR, data processing, and machine learning into a real-world application.

This project enhanced my understanding of data engineering and AI workflows.